
Agency in action selection and action execution produce distinct biases in decision making

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Abstract

Our sense of agency influences our decisions. The extent to which we have agency depends on whether we are free to select actions and whether our execution of those actions determines their outcomes. Past research suggests that free choice adds value—we prefer options that we freely selected compared to those we were forced to select. This selection bias also appears to be asymmetrical, increasing the value of successful options but not influencing the value of unsuccessful ones. We hypothesize that a different source of agency—control over the outcome of our actions—impacts how we perceive options linked to negative outcomes. In this study, we manipulated these two sources of agency to examine how they impact choice biases. We conducted three experiments in which the participants were either free to choose between two options or forced to select one, and where the outcome was either determined by a lottery or movement accuracy. During the training phase, we manipulated the reward probabilities for each option. To examine selection and execution biases, we presented paired options of identical value during a test phase. In the first experiment, we manipulated agency in action execution and found that the bias was restricted to outcomes associated with negative outcomes. In the second experiment, we manipulated agency in either selection or execution. With agency in action selection, participants exhibited a bias towards options associated with positive outcomes; in contrast, with agency in action execution, participants exhibited a bias towards options associated with negative outcomes. In the third experiment, we found that these two sources of bias were additive. Collectively, these findings demonstrate how different sources of agency give rise to distinct biases.

Keywords: Human decision making; choice biases; agency

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1 Introduction

Many decisions require that we not only select an action but also perform a movement in order to achieve the desired goal. After landing in the water following a long iron shot, a golfer must decide whether they selected the wrong club or executed a poor swing, with this inference impacting their choice when facing a similar shot in the future.

Humans combine incremental reward learning with cognitive biases and heuristics to guide their decisions. While expected value is a major determinant in these decisions, it is well established that other factors also influence our choice behavior. These include task goals [1, 2], motivational states [3], and various sources of bias [4]. These biases depend not only on the dynamics of the environment but also on our sense of agency or level of control within that environment. On the action selection front, we prefer options that we freely selected compared to options we were forced to select, even when the options have identical value. This bias appears to be asymmetric, occurring only when options are linked to positive outcomes [5]. In terms of agency in action execution—situations in which our movements influence the outcome of the selected action—biases appear to be stronger when the options are linked to negative outcomes [6]. It is unclear whether the same mechanism that selectively amplifies positive reward prediction errors related to action selection also selectively attenuates negative reward prediction errors related to action execution. Here, we report three experiments that explore the differential impacts of these two sources of agency.

2 Methods

We manipulated agency in action selection and action execution. To manipulate agency in action selection, we compared trials where participants were free to choose with trials where they had no choice (FC vs. NC). For agency in action execution, we compared trials where outcomes were determined by a lottery with trials where outcomes were based on movement accuracy (L vs. M). For lottery trials, participants pressed a button on a digitizing tablet (Wacom Co, Kazo, Japan) to indicate their choice. For movement trials, they held a pen mounted on a sliding disk and made reaching movements to indicate their choice, with the tablet recording the position of the pen. The stimuli were displayed on a horizontally mounted screen positioned above the tablet, which occluded vision of the hand (Figure 1A). The study protocol was approved by the Institutional Review Board at UC Berkeley.

Participants completed a training phase followed by a test phase. Six stimulus pairs were presented during the training phase. Three of these pairs were abstract symbols, and three were target locations evenly spaced around an invisible circle (Figure 1B). Each symbol/location was associated with probabilistic feedback, and within each pair, one more reliably led to positive feedback than the other. Participants completed four training blocks. Each block consisted of 120 trials where the six stimulus pairs were presented 20 times each in a pseudorandom order. Participants were then tested on different combinations of symbol pairs and target pairs, as well as the six choice bias pairs of symbols and targets with identical value.

Participants were sometimes forced to select a specific option. For these no choice trials, we highlighted the designated option with a yellow square. If participants did not select it, an ‘out of bounds’ message was displayed. The choice and outcome were yoked to previous free choice trials to match sampling and reinforcement histories for pairs with the same reward probabilities. Free choice and no choice trials were pseudorandomized, and the yoked pair was presented in the following sequence.

We informed participants that on some trials the reward outcomes would be determined by a lottery, and on other trials, the outcome would be based on reach accuracy. For reach accuracy trials, the participant moved their hand to position a cursor within a start circle positioned at the center of the screen. After this position was maintained for a randomized time between 600 ms and 1200 ms, two targets appeared. Once a reach was initiated, cursor feedback was removed until the reach passed the target, at which point participants received cursor feedback and monetary feedback (Figure 1C; top row). Participants were informed they would earn money only when the cursor intersected the target. Unbeknownst to them, the visual position of the cursor was sometimes perturbed in order to impose the desired reward probabilities. On rewarded trials, +1 cent appeared on the screen and was added to the total, while on unrewarded trials, -1 cent appeared and was subtracted from the total. To ensure participants were unaware of the manipulation of cursor feedback, we

displayed an ‘out of bounds’ message when a reach deviated more than 30° from the target. For lottery trials, a fixation cross appeared, and after a randomized delay between 600 ms and 1200 ms, the two symbols for that trial were displayed vertically in the center of the screen, with their positions randomly assigned to either the top or bottom. Participants pressed one of two buttons to indicate their choice, and after a 1000 ms delay, received monetary feedback. By adding this feedback delay, we approximately matched the total duration of the reach and lottery trials (Figure 1C; bottom row). Participants indicated their choices in the same way during the training and test phases—either by reaching for a target or pressing a key for a symbol—however, they did not receive feedback during the test phase.

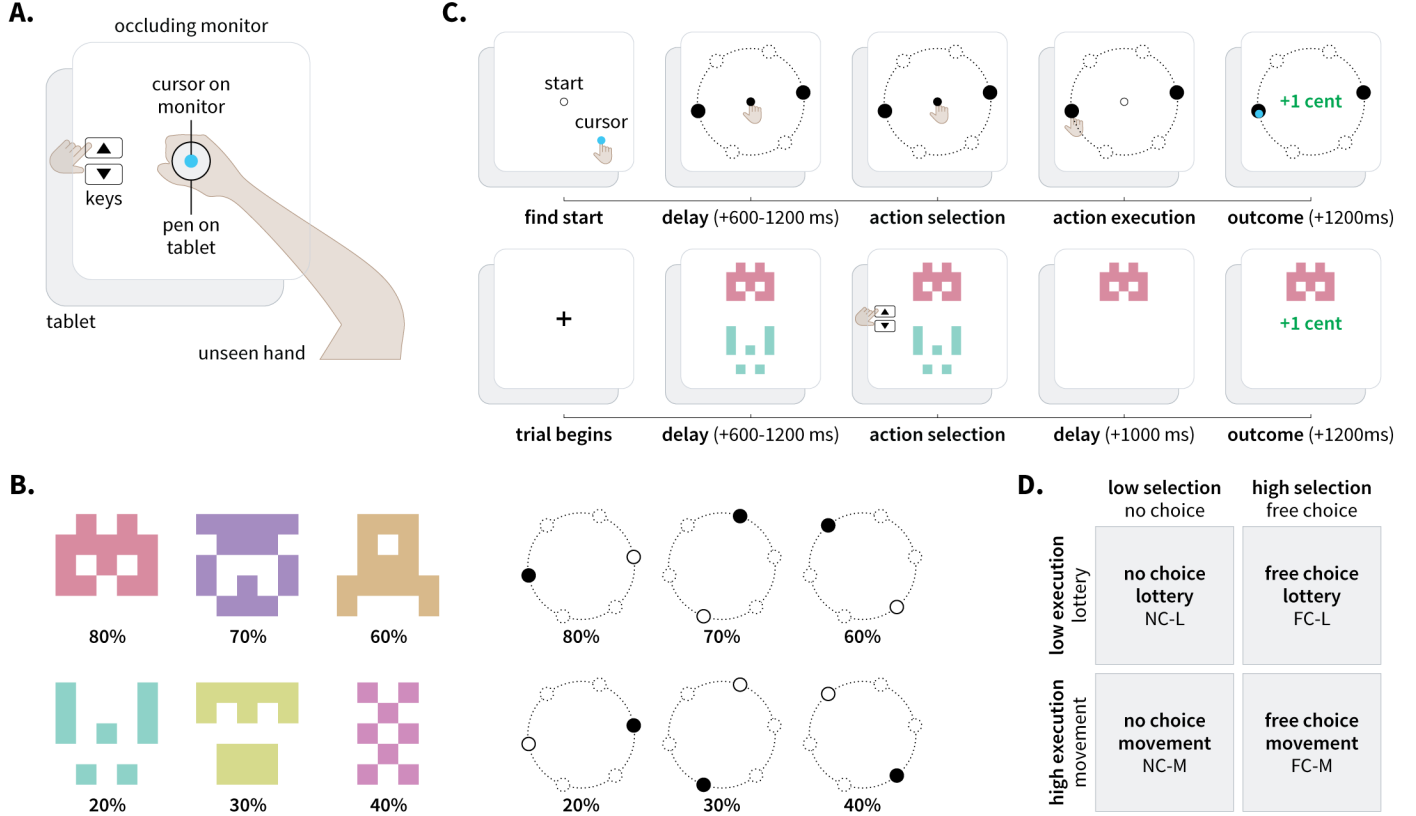


Figure 1. Experimental design. **A.** Participants indicated their choices on a tablet by pressing a key or reaching to a target location. **B.** Each column indicates a pair of stimuli (symbols or locations) used during the training phase, along with specific reward probabilities. **C.** Example of FC-M (top) and FC-L (bottom) training trials. For the former, participants selected which location to reach towards with the outcome indicated by monetary and cursor feedback. For the latter, participants selected a symbol by pressing one of two response keys with the outcome indicated by monetary feedback. **D.** 2x2 schematic illustrating the different conditions created by manipulating agency in selection and execution.

We manipulated agency in selection and execution and assessed patterns of bias associated with these two levels of control (Figure 1D). Three experiments differed in which combinations of agency were compared. In experiment 1 ($n = 21$, 14 female, age = 20.7 ± 2.7 yrs), we compared low execution agency (lottery) with high execution agency (movement), while keeping agency in selection constant (always free choice), to test for a negative bias (Figure 2A; light green). In experiment 2 ($n = 21$, 14 female, age = 20.6 ± 1.7 yrs), we compared agency in selection (free choice, lottery) and execution (no choice, movement) to again test for asymmetric biases. Based on [5] and experiment 1, we predicted that participants would exhibit biases for selected options associated with positive outcomes, whereas they would exhibit biases for executed options associated with negative outcomes (Figure 2A; light blue). In experiment 3 ($n = 21$, 16 female, age = 22.6 ± 3.7 yrs), we examined how these biases interact when agency is present for both selection and execution, comparing the lowest (no choice, lottery) and highest (free choice, movement) agency conditions (Figure 2A; dark green).

3 Results

3.1 Learning rates were similar across conditions involving different types of agency

We examined the rate at which participants' choices converged on the correct choice. We compared the time constant of an exponential curve fitted to the free choice trials in Experiment 1, where participants selected between lottery-based options and movement-based options. Across all pairs, the learning rate was 33.12 (95% CI [14.86, 51.38]) for FC-L trials and 32.58 (95% CI [13.39, 51.78]) for FC-M trials (Figure 2B; light green).

3.2 Agency in selection results in positive biases, while agency in execution results in negative biases

In Experiment 1, we manipulated agency only in action execution, as all options were learned through free choice, keeping agency in action selection constant (Figure 2C; light green). We used a linear mixed effects model to examine choice biases with options of equal value. The results showed minimal biases for options associated with positive outcomes (80%: $\beta = 22.62$, $p = 5.99 \times 10^{-4}$; 70%: $\beta = 2.38$, $p = 0.71$; 60%: $\beta = -6.55$, $p = 0.31$; combined: $\beta = 6.15$, $p = 0.15$). In contrast, options associated with negative outcomes were biased toward conditions of high agency in execution (movement) compared to low agency in execution (lottery) (40%: $\beta = -27.98$, $p = 2.76 \times 10^{-5}$; 30%: $\beta = -35.12$, $p = 2.45 \times 10^{-7}$; 20%: $\beta = -46.43$, $p = 4.74 \times 10^{-11}$; combined: $\beta = -36.51$, $p = 8.40 \times 10^{-11}$). Despite the objective value equivalence between these options, participants perceived the values differently depending on the source of the outcomes.

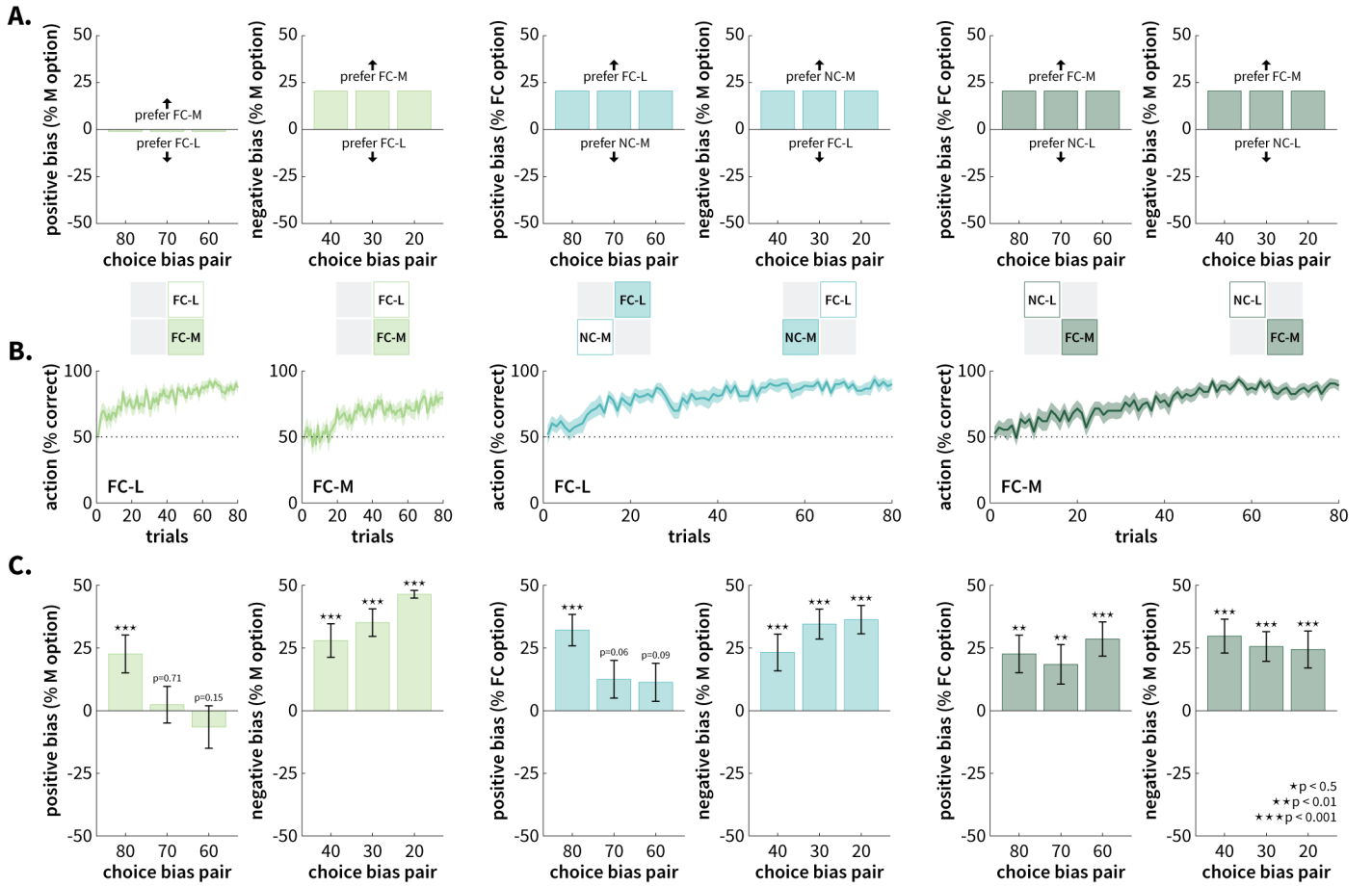


Figure 2. Behavioral results. **A.** Predicted biases in Experiment 1 (FC-L vs. FC-M; left panel), Experiment 2 (FC-L vs. NC-M; middle panel), and Experiment 3 (NC-L vs. FC-M; right panel). Within each panel, we display options associated with positive outcomes on the left and options associated with negative outcomes on the right. The bottom row displays the quadrant of conditions used during training (white and colored) for each experiment. The top row y-axis is based on the stimulus from the colored box condition. **B.** Performance changes during the training phase across all pairs in each of the three experiments. **C.** Positive and negative choice biases in Experiments 1–3.

Participants exhibited the predicted biases when we manipulated either action selection or action execution (Figure 2C; light blue). With options learned through free choice and lottery-based outcomes (FC-L), we observed a bias toward those associated with positive outcomes (80%: $\beta = 32.14$, $p = 3.41 \times 10^{-6}$; 70%: $\beta = 12.50$, $p = 0.061$; 60%: $\beta = 11.31$, $p = 0.089$; combined: $\beta = 18.65$, $p = 1.53 \times 10^{-3}$). With options learned through no choice and movement-based outcomes (NC-M), we instead observed a bias towards those associated with negative outcomes (40%: $\beta = -23.21$, $p = 6.14 \times 10^{-4}$; 30%: $\beta = -34.52$, $p = 7.19 \times 10^{-7}$; 20%: $\beta = -36.31$, $p = 2.15 \times 10^{-7}$; combined: $\beta = -31.35$, $p = 1.17 \times 10^{-6}$). The positive bias appears to be specific to conditions with agency in action selection, while the negative bias is specific to conditions with agency in action execution.

Participants exhibited the predicted biases when we manipulated agency in selection and execution for the same options (Figure 2C; dark green). Participants experienced the highest sense of agency for options learned through free choice and movement-based outcomes (FC-M) and exhibited a bias towards these options for positive (80%: $\beta = -22.62$, $p = 1.36 \times 10^{-3}$; 70%: $\beta = -18.45$, $p = 8.51 \times 10^{-3}$; 60%: $\beta = -28.57$, $p = 6.43 \times 10^{-5}$; combined: $\beta = -23.21$, $p = 2.16 \times 10^{-4}$) and negative (40%: $\beta = -29.76$, $p = 3.30 \times 10^{-5}$; 30%: $\beta = -25.60$, $p = 3.14 \times 10^{-4}$; 20%: $\beta = -24.40$, $p = 5.74 \times 10^{-4}$; combined: $\beta = -26.59$, $p = 3.48 \times 10^{-5}$) choice bias pairs. The presence of both positive and negative biases suggest that these different sources of agency are additive.

3.3 The influence of agency in action selection and action execution on reward sensitivity

One concern is that the observed biases may be due to fluctuations in attention either between lottery-based or movement-based trials, or between free choice and no choice trials. To address this, we evaluated choice rates for test options ranging from the lowest to highest reward probability. If participants accurately learned the reward probabilities, options with lower probabilities would have lower choice rates, while options with higher probabilities would have higher choice rates, resulting in a positive slope. When combining free choice and no choice options across experiments, we observed a positive slope for both lottery-based (0.99, 95% CI [0.88, 1.11], $p = 2.47 \times 10^{-49}$) and movement-based (0.73, 95% CI [0.60, 0.85], $p = 4.76 \times 10^{-27}$) options. We also observed a positive slope for free choice (1.01, 95% CI [0.91, 1.11], $p = 3.29 \times 10^{-68}$) and no choice (0.56, 95% CI [0.41, 0.71], $p = 1.15 \times 10^{-11}$) when combining lottery-based and movement-based options across experiments.

4 Discussion

Across three experiments, we found that different sources of agency gave rise to biases that were additive but differentially modulated by outcomes. Perhaps counterintuitively, some theories propose that biased strategies can be more effective than unbiased ones. For example, overconfidence—inflating the perceived value of our actions—can maximize individual fitness [7]. Our results provide behavioral insights into the basis of this overconfidence and its nuances. Returning to our golfer example, these results suggest that following a well-executed shot, their club selection is reinforced, while after an errant shot, the focus is on repeating the movement to improve swing execution.

References

- [1] Frömer, R., Dean Wolf, C. K., & Shenhav, A. (2019). Goal congruency dominates reward value in accounting for behavioral and neural correlates of value-based decision-making. *Nature communications*, 10(1), 4926.
- [2] Molinaro, G., & Collins, A. G. (2023). Intrinsic rewards explain context-sensitive valuation in reinforcement learning. *PLoS Biology*, 21(7), e3002201.
- [3] Howard, J. D., & Kahnt, T. (2017). Identity-specific reward representations in orbitofrontal cortex are modulated by selective devaluation. *Journal of Neuroscience*, 37(10), 2627-2638.
- [4] Bornstein, A. M., Khaw, M. W., Shohamy, D., & Daw, N. D. (2017). Reminders of past choices bias decisions for reward in humans. *Nature Communications*, 8(1), 15958.
- [5] Cockburn, J., Collins, A. G., & Frank, M. J. (2014). A reinforcement learning mechanism responsible for the valuation of free choice. *Neuron*, 83(3), 551-557.
- [6] McDougle, S. D., Boggess, M. J., Crossley, M. J., Parvin, D., Ivry, R. B., & Taylor, J. A. (2016). Credit assignment in movement-dependent reinforcement learning. *Proceedings of the National Academy of Sciences*, 113(24), 6797-6802.
- [7] Johnson, D. D., & Fowler, J. H. (2011). The evolution of overconfidence. *Nature*, 477(7364), 317-320.